

flyPad taxi route page

Ground > Taxi: the taxi route on the Navigraph airport map, from the gate to the runway holding point and from the runway exit to the gate, suggested and edited to the ATC clearance

Branch	pr/26-efb-taxi-route
Aircraft	A32NX and A380X (flyPad)
Commits	b97742c40 feat(efb): taxi route page on the Navigraph airport map; 2026-09-29: 6c2f022f6 fix(efb): taxi route with multi-word taxiways, entry without runway crossing, locate button
Base	PR 23 (landing calculator: the shared runway exit module btwExits.ts)
Status	CI checks pass; merged into develop 9cd78dd05; tested in the sim on 2026-09-28 at RJTT (user-confirmed).

Purpose

The A380 OANS shows the airport map but no taxi route: the 2011 A380 FCOM has no taxi routing (the crew marks the map with flags and crosses, DSC-34-10-70-20) and MSFS has no D-TAXI. The flyPad page gives the crew the route of the ATC taxi clearance on the same Navigraph airport map (AMDB) the OANS uses, with the runway crossings, so the cockpit displays stay true to the FCOM.

Specification

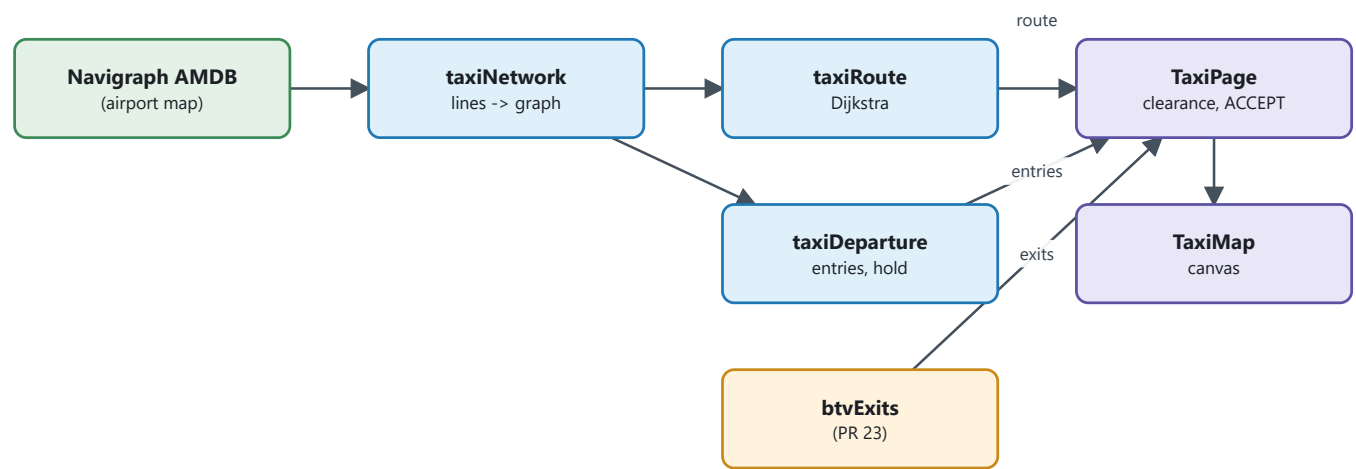
Item	Behaviour
Departure (default)	from a gate / ramp / cargo stand, or the aircraft position, to a runway entry (the first one = full length, the others with the runway length remaining); the route stops at the runway holding position line crossed last (HOLD RWY stop bar)
Arrival	from a runway exit (the exits of the landing calculator) or the aircraft position to a stand
Suggestion	the shortest route along the taxi lines; SUGGEST copies its taxiways into the clearance field
ATC clearance	the taxiways typed in order: the route follows them strictly (unnamed connecting lines allowed, the runway entry line after the last one); ACCEPT draws it solid; an impossible clearance is shown in red
Map	aprons, taxiways, runways, guidance lines, stands (tap one to choose it), taxiway names, runway entries, route with its taxiway names, runway crossings in red, aircraft symbol; drag, zoom, fit
Stand names	typed with or without leading zeros and separators ("16" finds "016"); suggestions list the names that start with, then contain, the typed text

Design

- Shared module `fbw-common/src/systems/shared/src/taxi/`: `taxiNetwork.ts` builds the network from the AMDB guidance lines (taxiway, runway exit and stand guidance lines; runway centrelines only split the lines that cross them and are never followed); lines split where they cross, line ends within 3 m joined (T junctions), a dead end of a disconnected part joined to the nearest line of another part within 15 m (gaps in the airport map).
- `taxiRoute.ts`: Dijkstra over (node, index of the cleared taxiway) states; unnamed lines cost x1.2; the lead-in lines of other stands and other exits x3 as a last resort; from the aircraft, a 2000 m penalty for starting against its heading. `taxiRouteRunwayCrossings` samples the route every 5 m in the runway polygons.
- `taxiDeparture.ts`: the runway entries (lines leaving the runway area within 40 m of the centreline, unnamed lead-on lines named after the taxiway they come from) and the cut at the holding point (last holding position line crossed within the last 400 m).
- `taxiAmdb.ts`: lines, stands, runways, runway axes, holding lines and map shapes from the AMDB features; stand name matching.

- flyPad: `Ground/Pages/Taxi` (TaxiAirport loads the AMDB layers through the Navigraph client, cache of 3 airports; TaxiMap canvas drawn larger than the view so a drag only moves it; TaxiPage), store slice `taxiRoute` (kept across the flyPad pages), Taxi tab in Ground.

Component added or changed by the feature MSFS / simulator Aircraft system (systems host, FMS, FWS) Display / page
Data, LVar, storage Crew, flypad, external program



Architecture

Tests

9 unit tests in `taxi/taxiRoute.spec.ts` on a synthetic airport: suggestion and clearance routes, clearance errors, runway crossings, the runway entries (full length, crossing taxiway, unnamed lead-on line), the holding point cut, an intersection departure, stand name matching, the join of a lead-in line that stops short of the taxi line; plus the runway exit start points in `btvExits.spec.ts`.

Limitations

- Needs a Navigraph account (the AMDB); without it the page says so.
- The route follows the guidance lines of the map; one-way taxiways, closures and NOTAMs are not known.
- The route is shown on the flyPad only, not on the OANS (not in the FCOM).

Fixes of the CYUL test and the locate button (2026-09-29)

Item	Behaviour
Taxiway names of several words	<code>parseTaxiClearance</code> reads the clearance with the names of the airport, the longest first (SOUTH RAMP, NORTH RAMP); other words are one taxiway each
Proposed runway entry	<code>taxiDefaultEntry</code> : the first entry from the threshold whose suggested route (cut at the holding point) crosses no runway (crossings within 40 m of the entry ignored); the full length entry when every route crosses one
Aircraft on the first cleared taxiway	the start is already along the clearance (it gave no route)
Locate button	centres the map on the aircraft, at 1 px/m at least (stand names); greyed away from the airport

PR description (from the PR draft)

From `pr-drafts/26-efb-taxi-route.md`, section "Summary of Changes" onwards.

Summary of Changes

A new flyPad page, Ground > Taxi (both aircraft), shows the taxi route on the Navigraph airport map (AMDB, the data of the A380 OANS). The A380 FCOM has no taxi routing (the crew marks the OANS with flags and crosses, DSC-34-10-70-20) and MSFS has no D-TAXI, so the route lives on the flyPad and the cockpit displays stay as the FCOM describes them.

- **Departure** (default): from a gate / ramp / cargo stand, or the aircraft position, to a runway entry. The entries of the selected runway are listed in order (the first one = full length, the others with the runway length remaining) and drawn on the map; the route stops at the runway holding position line (HOLD RWY stop bar).
- **Arrival**: from a runway exit (the exits of the landing calculator) or the aircraft position to a stand.
- **Suggestion and clearance**: the shortest route along the taxi lines is suggested (SUGGEST copies its taxiways into the clearance field); the crew types the taxiways of the ATC clearance in order, the route follows them strictly, ACCEPT draws it solid. Runway crossings are listed and circled in red; an impossible clearance or a start not connected to the end is explained.
- **Map**: aprons, taxiways, runways, guidance lines, taxiway names (spaced out, not overlapping), stands (tap one to choose it), runway entries, the route with its taxiway names, the aircraft; drag, zoom, fit. The canvas is drawn larger than the view, so a drag only moves it and the map is drawn again when the drag ends.
- **Stand names** are found with or without leading zeros and separators ("16" finds "016"); the suggestions list the names that start with, then contain, the typed text.
- Shared module `fbw-common/src/systems/shared/src/taxi/`: `taxiNetwork.ts` (graph from the guidance lines: split at crossings, T junctions within 3 m, dead ends of disconnected parts joined within 15 m; runway centrelines only split crossing lines and are never followed), `taxiRoute.ts` (Dijkstra over node x cleared-taxiway states; runway crossings), `taxiDeparture.ts` (runway entries, holding point cut), `taxiAmdb.ts` (AMDB features, stand name matching). `btvExits.ts` gives each exit its start point on the runway.
- flyPad: `Ground/Pages/Taxi` (TaxiAirport, TaxiMap, TaxiPage), store slice `taxiRoute` (kept across the flyPad pages), Taxi tab.
- **A380X OANS**: when the crew accepts a route, the flyPad sends green flags to the OANS (the crew flags of FCOM DSC-34-10-70-20, "to locate or mark a given point on the airport") at the start of each taxiway and at the gate or holding point (`OansControlEvents.oans_taxi_route_flags`). The OANC shows them when it displays that airport, replaces the previous route's flags, removes them when the route is no longer accepted and never touches the crew's own flags.
- From the aircraft position, the route never takes the line the aircraft is on whole (no U-turn through the aircraft); the fitted map keeps a 70 px margin for the labels at the ends of the route.
- 9 unit tests (`taxi/taxiRoute.spec.ts`, incl. the flag points and the no-U-turn start) + the exit start points in `btvExits.spec.ts`.
- **Taxiway names of several words** (SOUTH RAMP): the clearance is read with the names of the airport, the longest first.
- **Proposed runway entry**: the first entry from the threshold whose suggested route crosses no runway; the full length entry when every route crosses one (the crew chooses).
- **Locate button** (pin, under the zoom buttons): centres the map on the aircraft, zoomed in with the stand names; greyed when the aircraft is not at the airport.
- 11 unit tests (+ multi-word clearance, entry without crossing).

Cockpit API Changes

New OANS control event `oans_taxi_route_flags` (`{ icao, points: lat/long }`), flyPad -> OANC, synced.

Screenshots (if necessary)

TO ADD: a departure route from a gate to a runway holding point with a runway crossing; an arrival from an exit to a gate.

References

- A380 FCOM DSC-34-10-70-20 (Airport navigation): the OANS has no taxi routing (flags and crosses only), hence the flyPad page.
- EUROCAE ED-99 airport mapping database features used by the Navigraph AMDB (taxiway guidance lines, runway exit lines, stand guidance lines, parking stand locations, runway holding positions, painted centrelines).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. Navigraph account linked in the flyPad settings. Ground > Taxi, ORIG: the airport map loads.
2. Departure: choose a gate (type it or tap it on the map), a runway and an entry: a dashed route to the HOLD RWY bar appears; SUGGEST, then ACCEPT: the route turns solid.
3. Type another clearance (e.g. a longer taxiway sequence): the route follows it in order; an impossible one is shown in red.
4. From the aircraft: at the gate, Aircraft: the route starts at the aircraft.
5. Arrival after landing: runway, exit and gate: the route from the exit to the gate, with the runway crossings in red.
6. Drag and zoom the map: smooth; the taxiway names are shown.
7. `npx vitest run fbw-common/src/systems/shared/src/taxi` (9 tests).